A Comparative Evaluation of Mathematical Detection Models for Prompt Injection in Large Language Models

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Abstract— **Prompt injection attacks represent a critical security vulnerability in large language model systems, where adversarial crafted inputs cause the model to override its system-level instructions and execute unintended behavior. This paper presents a comparative evaluation of six mathematically grounded detection models across four distinct feature spaces: token probability, distributional, supervised probabilistic, and semantic embedding. The four base classifiers, Shannon Entropy, KL Divergence, Naïve Bayes with TF-IDF vectorization, and a Sentence Transformer Embedding classifier using all-MiniLM-L6-v2, are trained and evaluated independently on the neuralchemy/Prompt-Injection-Dataset. Two combined models are then evaluated: a Combined Early Fusion classifier that projects scalar features to a 32-dimensional representation before concatenating them with the 384-dimensional sentence embedding into a 416-dimensional joint input vector, and a Combined Late Fusion Stacking classifier that trains a logistic regression meta-model on the stacked probability outputs of all four base classifiers. Experimental results on a labeled dataset of 6,274 samples show that the Combined Early Fusion model achieves the best performance across all metrics, including macro-averaged F1 and ROC AUC. The results confirm that no single feature space is sufficient for reliable prompt injection detection, and that feature-level early fusion with scalar projection successfully integrates complementary signal across heterogeneous mathematical representations to achieve the strongest detection performance.**

Keywords— Prompt Injection, Large Language Models (LLMs), Adversarial Machine Learning, Binary Classification, Shannon Entropy, Kullback-Leibler divergence, Naïve Bayes, Sentence Transformers, Feature Fusion, Early Fusion, Late Fusion Stacking, Ensemble Learning, Neural Network

I. INTRODUCTION

The widespread development and deployment of large language models (LLMs) across enterprise and consumer applications have fundamentally altered the surface area of software security. Unlike conventional software systems, LLMs accept natural language as both instruction and data, making it architecturally impossible to enforce a strict separation between trusted control flow and untrusted user input at the token level. This structural property creates a class of vulnerability known as prompt injection, where in an adversary embeds instructions within user-supplied or externally retrieved content that cause the model to override its system-level directives, exfiltrate sensitive context, or execute unintended actions.[1] As of 2025, prompt injection remains ranked first on the OWASP Top 10 for Large Language Model Applications, a position it has held continuously since the list's inception, reflecting the persistence and severity of the threat across deployed systems.[2]

Prompt injection attacks manifest in two principal forms. Direct injection occurs when a user creates input that explicitly overrides system instructions, typically through imperative phrasing such as "Ignore all previous instructions." Indirect injection, identified by as the more operationally significant variant in production deployments, occurs when adversarial instructions are embedded within external content, such as retrieved documents, web pages, tool outputs, or email, that the model processes autonomously and without user awareness.[1] A comprehensive review synthesizing 128 peer-reviewed studies published between 2022 and 2025 found that sophisticated multimodal injection attacks achieve success rates exceeding 90% against unprotected systems, while existing input preprocessing defenses achieve only 60-80% detection rates against known attack patterns, with substantially lower coverage against novel vectors.[3]

The fundamental limitation of current defenses is their reliance on single-mechanism detection. Blocklist filtering, perplexity thresholding, and output monitoring each operate within a single representational space, such as surface token patterns, statistical token distributions, or semantic embeddings, and an adversary with knowledge of that space can engineer evasive inputs accordingly. Lan et al. (2025) demonstrated this limitation empirically, showing that combining banned-term filtering with embedding-based similarity search and a transformer classifier achieved meaningfully higher detection coverage than any individual component in isolation.[4] Similarly, Jayathilake (2025) demonstrated that sentence embedding-based classification using federated logistic regression achieves robust detection performance across distributed deployments while preserving data privacy, establishing that semantic embedding representations carry strong discriminative signal for injection detection that complements surface-level statistical approaches.[5]

This paper presents a comparative evaluation of six mathematically grounded detection models for prompt injection and proposes two combined model architectures that integrate the four base feature representations through distinct fusion strategies. The four base models span four distinct representational spaces: (1) token probability space, addressed through Shannon entropy analysis, which measures information-theoretic anomalies in the character frequency distribution of an input; (2) distributional space, addressed through Kullback-Leibler (KL) divergence, which quantifies the statistical distance between the input's token distribution and a reference benign distribution; (3) supervised probabilistic classification space, addressed through a Naïve Bayes classifier trained on labeled injection and benign corpora using TF-IDF vectorization; and (4) semantic embedding space, addressed through 384-dimensional sentence-level embeddings produced by the all-MiniLM-L6-v2 sentence transformer encoder. The fifth model, the Combined Early Fusion classifier, projects the three scalar base features to a 32-dimensional representation before concatenating them with the 384-dimensional sentence embedding, forming a 416-dimensional joint input vector trained end-to-end on a unified neural network classifier. This feature-level early fusion architecture is theoretically grounded in the two-class fusion literature: early fusion has been shown to achieve the Bayes-optimal error rate under full model knowledge and to converge to this optimum as training set size increases.[6] The sixth model, the Combined Late Fusion Stacking classifier, trains a logistic regression meta-model on the stacked probability outputs of all four base classifiers, learning an optimal weighted combination of their individual decisions. The two fusion strategies represent complementary design philosophies, early fusion enables cross-space interaction learning at the feature level, while late fusion stacking is computationally simpler and interpretable through the meta-model's learned coefficients.

Experimental evaluation on the neuralchemy/Prompt-Injection-Dataset yields the following macro-averaged F1 scores across all six models: Shannon Entropy (83.51%), KL Divergence (77.38%), Naïve Bayes (93.66%), Sentence Transformer Embedding (94.09%), Combined Early Fusion (95.18%), and Combined Late Fusion Stacking (94.64%). These results support three principal findings. First, models operating in semantic embedding space substantially outperform those operating in token probability space, the gap between KL Divergence (77.38%) and the embedding model (94.09%) represents a 16.71 percentage point difference, indicating that surface-level statistical anomaly detection is insufficient against semantically coherent injection payloads that conform to normal token distributions. Second, both combined models outperform the other four individual base classifiers (Late Fusion Stacking's ROC AUC score occasionally outperforms Embedding's and vice vera) across the reported metrics, confirming that integrating heterogeneous feature representations captures complementary discriminative signal unavailable to any single-space classifier. Third, the Combined Early Fusion classifier achieves the highest macro F1 of 95.18% and the highest ROC AUC of 99.36% across all six models, surpassing the Combined Late Fusion Stacking classifier (macro F1: 94.64%, ROC AUC: 99.06%) and demonstrating that feature-level fusion with scalar projection outperforms decision-level stacking when cross-space complementarity is strong and scale normalization is applied correctly.

The rest of this paper is organized as follows. Section II reviews related work in prompt injection defense and feature fusion methods. Section III formalizes the threat model and defines the detection problem. Section IV presents the mathematical formulations of each of the six detection models. Sections V describe the combined models' early fusion and late fusion stacking strategy. Section VI presents the experimental methodology and comparative results. Section VII discusses implications, limitations, and directions for future work, and Section VIII concludes.

II. Related Work

Prompt injections occur when a user crafts an input that causes an LLM to behave in ways it was not intended to, including bypassing safety guidelines, leaking sensitive information, or taking unauthorized actions. A key challenge is that these inputs do not need to be human-readable, if the model processes the text, the attack can take effect. Even commonly used techniques such as RAG and fine-tuning have not been shown to fully prevent prompt injection.[2]

Lan et al. (2025) propose a real-time detection system that runs three mechanisms in parallel: a banned-term filter, an embedding similarity search, and a BERT-based classifier. The system screens input before they reach the model and flags those that match known injection patterns or sit close to known malicious examples in embedding space. When tested on a question answering dataset of over 13,000 records, the system reached an accuracy of approximately 0.98.[4] Their approach shows that using multiple detection signals together produces stronger results than relying on any one method alone, which is a core motivation for the multi-model design in this paper.

Shannon (1948) introduced entropy to measure information content in a probability distribution, and it has since been used to detect statistically unusual patterns in text inputs. KL divergence extends this idea by measuring how different one probability distribution is from another, making it useful for flagging inputs whose token distributions differ significantly from a known benign baseline.[8] Naive Bayes uses Bayes' theorem with a conditional independence assumption to classify text based on term probabilities, and has long been used as a reliable and lightweight text classifier.[9] Reimers and Gurevych (2019) introduced sentence transformer embeddings, which encode the full meaning of a sentence into a dense vector, making them effective at capturing semantic content that surface-level statistical features miss.[10] Jayathilaka (2025) applied sentence embeddings directly to prompt injection detection using federated logistic regression and showed they provide strong detection performance even in privacy-preserving distributed settings.[5] Scikit-learn provides the implementations of MultinomialNB, StandardScaler, and LogisticRegression used in this paper.[7]

Pereira et al. (2023) formally compare early fusion and late fusion for binary classification tasks. In early fusion, feature vectors from different sources are concatenated into one combined vector that is fed into a single classifier. In late fusion, each source has its own classifier, and the individual output scores are combined afterward. Their mathematical analysis shows that early fusion performs best when the model is fully known, but as training data becomes finite, early fusion degrades more than late fusion because the combined feature space grows larger and becomes harder to learn from. Late fusion is therefore sometimes the better practical choice when training data is limited. This paper applies both strategies to prompt injection detection across four different feature types, and the early fusion model's fast overfitting observed in the loss curves is consistent with this finding.[6]

III. Threat Model & Detection Problem

This section formalizes the security assumptions underlying the proposed detection framework and defines the prompt injection detection problem as a supervised binary classification task. The proposed system is a fixed ensemble detector composed of six purpose-built classifiers: a Shannon entropy neural network, a KL divergence neural network, a Naïve Bayes classifier with TF-IDF vectorization, a sentence transformer embedding classifier using all-MiniLM-L6-v2, a Combined Early Fusion classifier, and a Combined Late Fusion Stacking classifier. The two combined classifiers serve as the ensemble components, each integrating the four base feature representations through distinct fusion strategies for improved detection performance. All six classifiers are trained and evaluated on the neuralchemy/Prompt-Injection-Dataset sourced from HuggingFace.

A. System Model

The detection system is implemented as a fixed pipeline of six classifiers, each operating on a distinct mathematical representation of the raw input text and producing a continuous injection probability score in the range [0, 1]. All six classifiers share a common input — a raw natural language string — and are trained and evaluated independently. No classifier receives the output of another as its input during training, except for the Late Fusion Stacking model, which is trained on the validation-split probability outputs of the four base classifiers.

The feature extraction is handled by two components operating in parallel. The *EntropyKLFeatureExtractor* processes raw text by first computing the Shannon entropy of the character frequency distribution, then applying a *CountVectorizer*, fitted exclusively on benign training text, to produce a token count vector that is L1-normalized and compared against the learned benign reference distribution via KL divergence. The extractor additionally computes a special character ratio and a raw text length value, yielding a four-element feature vector of the form [entropy, KL, special character ratio, text length] for each input. Of these four values, only the two, entropy and KL, are used as inputs to their respective neural net classifiers. The special character ratio and text length are computed by not used by any classifier in the current system. In parallel, the all-MiniLM-L6-v2 sentence transformer encodes the raw text into a 384-dimensional dense embedding vector independently of the EntropyKLFeatureExtractor.

The four base neural network classifiers, Shannon entropy, KL divergence, embedding, and the Combined Early Fusion model, share the same feedforward architecture implemented in the EntropyClassifier class, consisting of three fully connected layers with dimensions input → 16 → 8 → 1, with ReLU activations and dropout regularization at rates of 0.3 and 0.15 respectively. The entropy and KL classifiers each receive a single scalar feature (input_dim = 1), and the embedding classifier receives the full 384-dimensional sentence vector (input_dim = 384). The Naïve Bayes classifier uses scikit-learn's MultinomialNB with TF-IDF vectorization capped at a vocabulary of 5,000 features and outputs a direct class probability without a sigmoid transformation.

The Combined Early Fusion classifier uses a distinct architecture, CombinedModelEarlyFusion, that receives a single concatenated tensor of [emb(384) | entropy(1) | kl(1) | nb(1)]. To prevent the 384-dimensional embedding from numerically dominating the three scalar features, the scalar features are first projected to a 32-dimensional representation via a dedicated linear layer with ReLU activation, then concatenated with the embedding vector to form a 416-dimensional joint input, which is passed through a classifier of dimensions 416 → 64 → 32 → 1 with ReLU activations and dropout.

The Combined Late Fusion Stacking classifier uses logistic regression as a meta-model trained on the stacked probability outputs of the four base classifiers. Each base model is run on the validation split to produce a (941 × 4) matrix of probability scores. The logistic regression meta-model is trained on this matrix against the true validation labels, learning a weighted linear combination of the four base model decisions. At test time, the four base models produce probability outputs for each input, which are stacked into a (942 × 4) matrix and passed to the meta-model for a final prediction.

During training, each of the four features, entropy, KL, embedding vectors, and Naïve Bayes probabilities, are independently standardized using StandardScaler fitted only on the training split, preventing data leakage across splits. The neuralchemy dataset provides pre-defined train, validation, and test splits, and all scalers fit exclusively on training data and applied without refitting to the validation and test splits. All four models are trained using the Adam optimizer with a learning rate of 1x10^-3, binary cross-entropy with logits loss, a batch size of 32, a maximum of 100 epochs, and early stopping with a patience of 5 epochs based on validation loss, with the best-performing checkpoint restored after training.

B. Attacker Model

We assume a black-box adversary who interacts with the deployed system solely through the natural language input interface and observes only the application's text outputs. The attacker has no knowledge of the detection pipeline, its component architecture, trained weights, decision threshold, or the benign reference distribution used by the KL divergence model.

C. Detection Problem

The prompt injection detection task is formulated as supervised binary classification. Each input is assigned a label of 1 for a prompt injection attempt or 0 for a benign input. Each of the six classifiers produces a continuous injection probability score in the range [0, 1], thresholder at 0.5 to yield a binary prediction. All six classifiers are trained and evaluated independently, and their results are compared directly with no single classifier is designated as a final decision maker.

Model performance across all six classifiers is evaluated using the macro-average F1 score. It is defined as the unweighted mean of the per-class F1 scores across both the injection and benign classes. It is calculated as **[7]**:

$$Macro\ F1 = \frac{1}{|N|}\sum_{k\epsilon|N|} 2 * \frac{TP}{TP + FP + FN} \tag{1}$$

where ***N*** is the number of classes, ***TP*** is the number of true positives, ***FP*** is the number of false positives, and ***FN*** is the number of false negatives.

It can also be expressed in terms of ***precision*** and ***recall***. It is calculated as:

$$Macro\ F1 = \frac{1}{|N|}\sum_{k\epsilon|N|} 2 * \frac{Precision * Recall}{Precision + Recall} \tag{2}$$

where ***precision*** is the ratio of true positives to all predicted positives (TP / (TP + FP)) and ***recall*** (sensitivity) is the ratio of true positives to all actual positives (TP / (TP + FN)).

Macro-averaging is selected to assign equal weight to both classes regardless of class imbalance in the evaluation dataset, providing a balanced and conservative performance measure that penalizes detection failures on either class equally.

D. Dataset

All six classifiers are trained and evaluated on the neuralchemy/Prompt-Injection-Dataset, a publicly available labeled corpus sourced from Hugging Face. The dataset provides pre-defined train, validation, and test splits used directly by the pipeline without any modifications. The dataset contains examples of direct injection attempts, jailbreak, system extraction, encoding obfuscation, persona replacement, indirect injection, token smuggling, many shot, crescendo, prompt leaking, context overflow, and benign. These are drawn from sources, NeurAlchemy original attack_db, HackAPrompt competition, WildGuard / JudgeComparison, HarmBench behavior goals, HarmBench benign counterparts, and Hand-crafted hard-negative prompts. The dataset is public, so it ensures full reproducibility of all experimental results. Complete statistics on dataset size, class distribution, and split sizes are provided in Section VI.

IV. Mathematical Forumulation

This section presents the mathematical specifications of each of the five detection models. For each model, the input representation, core mathematical operations, and output score are defined in the order they appear in the implementation. All models are operating on a common input, a raw natural language string, and produce an injection probability score $S \epsilon [0, 1]$ via sigmoid activation applied to a raw logit output, except for the Naïve Bayes classifier which outputs a direct class probability.

A. Shannon Entropy Analysis

The Shannon entropy model measures the information content of the character-level frequency distribution of the input string. Given input text ***T***, the relative frequency of each unique character ***c***, in the observed character set $\sum$ is computed as:

$$prob(c) = \frac{count(c, T)}{|T|}, c\epsilon\sum \tag{3}$$

where |T| is the total character length of T, defined as zero for an empty string. The Shannon entropy of the input is then computed as **[8]**:

$$H(T) = -\sum_{c\epsilon\sum} prob(c) * \log_2 prob(c) \tag{4}$$

The scalar H(T) is the sole input feature to the feedforward neural network f_H (input_dim = 1). Injection attempts exhibit characteristic entropy signatures: repetitive override phrases produce abnormally low entropy due to limited character diversity, while high-density adversarial high entropy. The sigmoid-activated output score is $S_H = \sigma\big(f_H\left(H(T)\right)\big)\epsilon[0, 1]$.

B. Kullback-Leibler Divergence

The KL divergence model measures how far the token distribution of the input departs from a reference distribution learned exclusively from benign training texts. A CountVectorizer is fitted on the benign training examples only. The mean token count vector across all benign texts is normalized by the total count sum to produce the benign reference distribution ***q***:

$$q = \frac{mean\_count\_benign}{\sum mean\_count\_benign + \varepsilon}, \varepsilon = 1 * 10^{-10} \tag{5}$$

At inference the input is transformed using the same fitted CountVectorizer and L1-normalized to yield the empirical input distribution:

$$p(T) = \frac{CountVec(T)}{\sum CountVec(T) + \varepsilon} \tag{6}$$

The KL divergence from the benign reference is then computed:

$$D_{KL}(p(T)|q) = \sum_i (p(i) + \varepsilon) * log\frac{p(i) + \varepsilon}{q(i) + \varepsilon} \tag{7}$$

where ε is added to both distributions to avoid undefined values at zero-probability tokens. The scalar D_{KL} is the sole input to the neural network f_{KL} (input_dim = 1), yielding the score $S_{KL} = \sigma(f_{KL}\left(D_{KL}\right))\epsilon[0, 1]$.

C. Naïve Bayes Classification

The Naïve Bayes model applies Bayes' theorem under the conditional independence assumption to a TF-IDF representation of the input. A TfidfVectorizer is fitted on the training corpus with vocabulary capped at 5,000 features and English stop words removed. The TF-IDF weight for term ***t*** in document ***d*** is:

$$tfidf(t, d) = tf(t, d) * \log\left(\frac{N}{df(t)}\right) \tag{8}$$

where $tf(t, d)$ is the term frequency of ***t*** in ***d, N*** is the total number of training documents, and $df(t)$ is the number of training documents containing ***t***. Under the conditional independence assumption, the posterior probability of the injection class is **[9]**:

$$P(y = 1 \mid d) = P(y = 1) * P(t_1 \mid y = 1) * P(t_2 \mid y = 1) * \dots * P(t_n \mid y = 1) \quad (9)$$

where $t1, t2, \dots, tn$ are the terms present in document $\boldsymbol{d}$. The classifier is implemented using scikit-learn's MultinomialNB fitted on the labeled training split. The posterior probability $P(y = 1 \mid d)$ is used directly as the injection score $s_{NB} \in [0, 1]$ without an additional sigmoid transformation, as MultinomialNB produces class probability natively.

D. Embedding Sentence Transformer

The embedding model maps each input string to a dense 384-dimensional semantic vector using the all-MiniLM-L6-v2 sentence transformer, a distilled MiniLM architecture pre-trained for semantic similarity task.**[10]**

$$E(d) = SentenceTransformer(\text{all-MiniLM-L6-v2})(d) \epsilon R^{384} \quad (10)$$

The embedding vector is independently standardized using a dedicated *StandardScaler* fitted on the training split and passed to the feedforward neural network f_E (input_dim = 384) sharing the same EntropyClassifier architecture. The embedding classifier captures semantic injection patterns that evade surface-level statistical detection. The adversarial inputs conforming to normal token distributions may still occupy anomalous regions of semantic embedding space, making this model complementary to the entropy and KL divergence approaches. The output is $S_E = \sigma\left(f_E\,(E(X))\right) \epsilon [0, 1]$.

E. Combined Early Fusion Model

The combined model implements early fusion feature concatenation, assembling the independently standardized feature representations of all four base approaches into a single 387-dimensional input vector prior to classification. The model is trained independently on this feature vector, so it does not receive the output scores of the other four classifiers. To prevent the 384-dimensional embedding from numerically dominating the three scalar features, the scalar features, standardized entropy, KL divergence, and Naïve Bayes probability, are first projected to a 32-dimensional representation via a dedicated linear layer with ReLU activation before concatenation:

$$x_{scalar_{proj}} = ReLU(W_{proj} * [z_H; z_{KL}; z_{NB}] + b_{proj}) \epsilon R^{32} \quad (11)$$

The projected scalar representation is then concatenated with the standardized embedding vector to form the 416-dimensional joint input:

$$z(d) = [z_E; s\_scalar\_proj] \epsilon R^{416} \quad (12)$$

where $z_E \in \mathbb{R}^{384}$ is the standardized sentence embedding and $x\_scalar_{proj} \in \mathbb{R}^{32}$ is the projected scalar representation. Each scalar feature is standardized to zero mean and unit variance prior to projection:

$$z_i = \frac{v_i - \mu_i}{\sigma_i} \quad (13)$$

where μ_i and σ_i are the mean and standard deviation of feature i computed over the training split only, ensuring no data leakage from the validation or test splits. The 416-dimensional vector *z(d)* is passed to a classifier of dimensions 416 → 64 → 32 → 1 with ReLU activations and dropout. The model is trained end-to-end using binary cross-entropy with logits loss:[11]

$$f_{BCE} = -\left(\frac{1}{N}\right)\sum_i [y_i * log\sigma(\hat{y}_i) + (1 - y_i) * \log(1 - \sigma(\hat{y}_i))] \quad (12)$$

where $y_i \in \{0,1\}$ is the ground truth label and $\hat{y}_i = f_C(z(d_i))$ is the raw logit output for sample $\boldsymbol{i}$, to which a sigmoid activation is applied at inference time to produce the final injection probability score $s_C = \sigma(f_C(z(d))) \in [0,1]$. By training across token probability, distributional, probabilistic, and semantic embedding spaces simultaneously with scalar projection ensuring balanced feature contribution, the Combined Early Fusion model achieves the highest macro F1 score of 0.9506 and the highest ROC AUC of 0.9932 among all six evaluated models.

F. Combined Late Fusion Stacking Model

The Combined Late Fusion Stacking model trains a logistic regression meta-model on the stacked probability outputs of all four independently trained base classifiers. Unlike the early fusion approach which operates on raw feature representations, late fusion stacking operates exclusively on the decision-level outputs of the base models, learning an optimal linear weighting of their individual injection probability scores.

After the four base classifiers are trained, each is run on the validation split to produce a probability score for every validation sample. These scores are stacked into a meta-feature matrix:

$$M_{val} = [\, s_H;\, s_{KL};\, s_{NB};\, s_E] \in \mathbb{R}\text{^}\{n_{val} \times 4\} \quad (13)$$

where $s\_H, s\_KL, s\_NB, s\_E \in [0,1]$ are the injection probability scores of the entropy, KL divergence, Naïve Bayes, and embedding classifiers respectively, and $n\_val = 941$ is the number of validation samples. The validation split is used rather than the training split to prevent data leakage, the base models have already seen the training data, so using training-split probabilities to train the meta-model would produce inflated estimates. A logistic regression meta-model is trained on M_val against the true validation labels, learning the decision function:

$$P(\, y = 1 \mid M\,) = \sigma(\, w^T \cdot M + b\,) \quad (14)$$

where $w \in \mathbb{R}^4$ is the learned coefficient vector assigning a weight to each base classifier's probability output and b is the bias term. At test time, the four base classifiers produce probability outputs for each test input, which are stacked into a (942 × 4) meta-feature matrix and passed to the trained logistic regression meta-model for a final injection probability score $s_{LF} = P(y = 1 \mid M_{test}) \in [0,1]$, thresholder at 0.5 for binary prediction. The meta-model's learned coefficient $vector\, w$ provides direct interpretability, a higher coefficient for a base classifier indicates that its probability output is assigned greater influence in the final stacking decision. The Combined Late Fusion Stacking model achieves a macro F1 of 94.64% outperforming all four individual base classifiers while ranking second among all six evaluated models. It also scored a ROC AUC of 99.06%, making it ranked third behind Embeddings at 99.13%.

V. Combined Model Strategies

The four base classifiers each operate within a distinct mathematical space, token probability, distributional, supervised probabilistic, and semantic embedding, and are trained independently. No single space is sufficient to detect all injection patterns, as an adversarial input can simultaneously appear normal in one space while being anomalous in another. This section describes the two combined model strategies evaluated in this paper: early fusion and late fusion stacking for the combined models.

Feature fusion strategies fall into two broad categories. In early fusion, raw feature representations from each source are concatenated into a single joint vector before any classification step, and a unified classifier is trained end-to-end on the combined representation. In late fusion stacking, each base classifier is first trained independently, then runs on a held-out split to produce probability outputs, and a meta-model is trained on those stacked probabilities to learn the optimal weighting of each base classifier's decision.

The early fusion model concatenates the independently standardized sentence embedding vector, entropy scalar, KL divergence scalar, and Naive Bayes probability scalar into a single input. To prevent the 384-dimensional embedding from numerically dominating the three scalar features, the scalar features are projected to a 32-dimensional representation before concatenation, yielding a 416-dimensional joint input vector. A unified neural network classifier is then trained end-to-end on this joint representation. Early fusion has been shown to achieve the Bayes-optimal error rate under full model knowledge for the two-class problem, though this advantage diminishes with finite training sets due to increased feature dimensionality.[6] Its principal advantage is that it allows the classifier to discover cross-space interaction effects between feature types that operating on output probabilities alone cannot capture.

The late fusion stacking model takes a complementary approach. Each of the four base classifiers, Shannon entropy, KL divergence, sentence transformer embedding, and Naive Bayes, is run on the validation split to produce a (941 × 4) matrix of probability scores. A logistic regression meta-model is trained on this matrix against the true validation labels, learning a weighted combination of the four base model decisions. At test time, the four base models produce probability outputs for each test input, which are stacked and passed to the meta-model for a final prediction. The meta-model is trained on the validation split rather than the training split to prevent data leakage from inputs the base models have already seen during training.

The two strategies represent different design philosophies. Early fusion preserves the internal structure of all feature representations and enables cross-space interaction learning, at the cost of higher input dimensionality and sensitivity to feature scale differences. Late fusion stacking is computationally simpler, interpretable through the meta-model's coefficients, and robust to the scale heterogeneity problem, but is limited to the information already summarized in each base model's output probability. Both strategies are evaluated experimentally in Section VI, and their performance is compared against the four individual base classifiers.

VI. Experiments

A. Dataset Statistics

All six classifiers are trained and evaluated on the neuralchemy/Prompt-Injection-Dataset sourced from Hugging Face. The dataset contains 6,274 labeled samples in total, of which 3,736 (59.5%) are prompt injection attempts (class 1) and 2,538 (40.5%) are benign inputs (class 0), reflecting a mild class imbalance of approximately 60/40. The dataset provides three pre-defined splits used directly without modification. Table I presents the full breakdown of each split by class.

TABLE I. Dataset Split Statistics

Split	Benign (0)	Injection (1)	Injection %	Total
Train	1,741	2,650	60.3%	4,391
Validation	407	534	56.8%	941
Test	390	552	58.6%	942
Total	2,538	3,736	69.5%	6,274

Table 1. The injection class is the majority class across all three splits. The mild imbalance of approximately 60/40 motivates the use of macro-averaged F1 as the primary evaluation metric to ensure equal weight is assigned to both classes.

TABLE II. Model Evaluation Results for All Five Classifiers on Test Split (N = 942)

Model	ROC AUC	F1 Benign	F1 Injection	F1 Macro	F1 Weighted
Shannon Entropy	90.66%	82.01%	85.02%	83.51%	83.77%
KL Divergence	85.85%	76.82%	77.93%	77.38%	77.47%
Naïve Bayes	98.39%	92.31%	95.01%	93.66%	93.89%
Embedding (all-MiniLM-L6-V2)	99.13%	93.08%	95.11%	94.09%	94.27%
Combined Early Fusion	**99.36%**	**94.32%**	**96.04%**	**95.18%**	**95.32%**
Combined Late Fusion Stacking	99.06%	93.73%	95.56%	94.64%	94.80%

Table 2. Table presents the ROC AUC, per-class (Benign & Injection) F1, macro-average F1, and weighted F1 scores for all six classifiers evaluated on the held-out test split of 942 samples (390 benign, 552 injection). Bolded numbers represent the best performance.

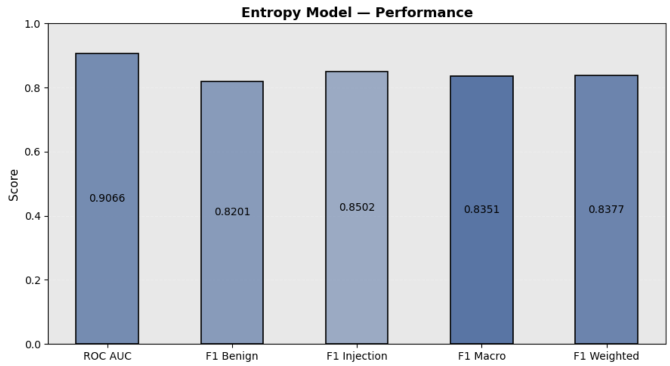

Fig. 1. Bar graph of Shannon Entropy Model's performance comparsion over five metrics ROC-AUC scores, F1 Benign, F1 Injection, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

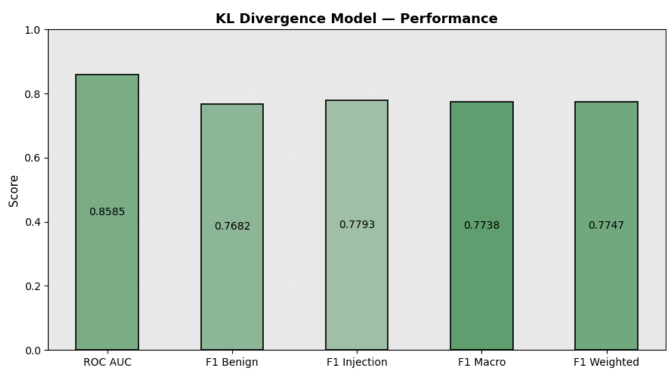

Fig. 2. Bar Graph of KL Divergence Model's performance comparsion over five metrics ROC-AUC scores, F1 Benign, F1 Injection, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

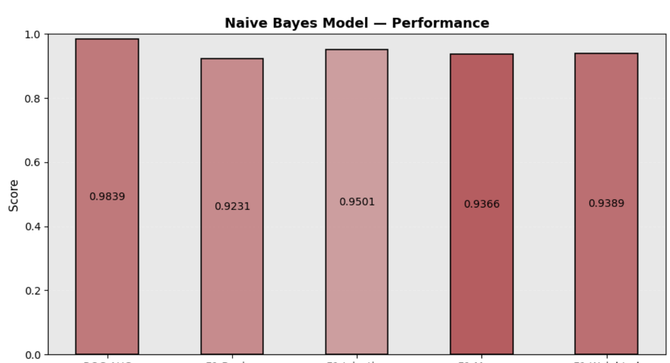

Fig. 3. Bar Graph of Naïve Bayes Model's performance comparsion over five metrics ROC-AUC scores, F1 Benign, F1 Injection, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

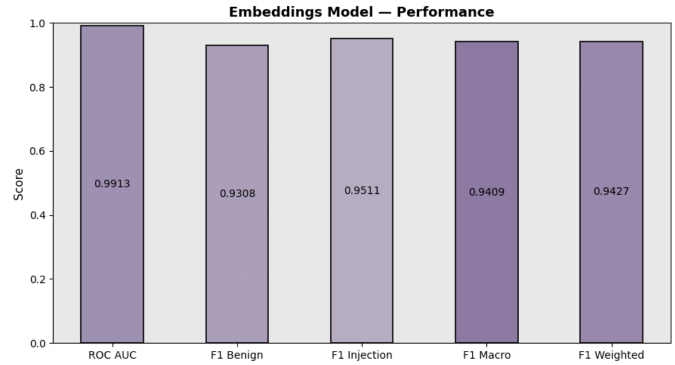

Fig. 4. Bar Graph of Embeddings Model's performance comparsion over five metrics ROC-AUC scores, F1 Benign, F1 Injection, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

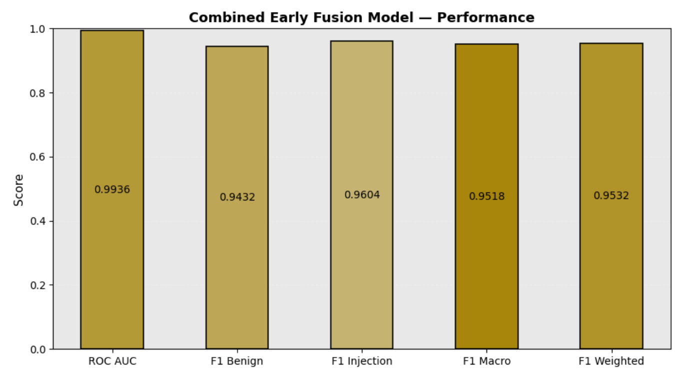

Fig. 5. Bar Graph of Combined Early Fusion Model's performance comparsion over five metrics ROC-AUC scores, F1 Benign, F1 Injection, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

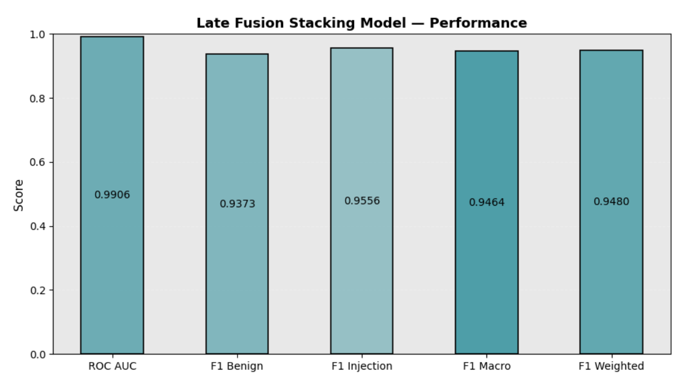

Fig. 6. Bar Graph of Combined Late Fusion Model's performance comparsion over five metrics ROC-AUC scores, F1 Benign, F1 Injection, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

B. Performance Analysis

The results reveal several findings worth discussing. First, the two information-theoretic models, Shannon entropy and KL divergence, perform substantially below the four supervised models, with macro F1 scores of 83.51% and 77.38% respectively. The gap of 16.71 percentage points between KL divergence (77.38%) and the best individual classifier indicates that character-level and token-distributional features alone are insufficient to reliably distinguish injection attempts from benign inputs, particularly against semantically coherent adversarial inputs that conform to normal statistical distributions. The KL divergence model's benign class F1 of 77.38% is the lowest recorded across all models and all metrics, reflecting a high false positive rate on benign inputs whose vocabulary deviates from the benign reference distribution without being adversarial.

Second, the Naïve Bayes classifier achieves a macro F1 of 93.66% and a ROC AUC of 98.39%, demonstrating that supervised pattern matching over a TF-IDF vocabulary is a strong baseline for prompt injection detection. Its injection class F1 of 95.01% exceeds that of both information-theoretic models by a wide margin, confirming that lexical features carry substantially more discriminative signal than distributional statistics alone for this task.

Third, the embedding model achieves a macro F1 of 94.09% and a ROC AUC of 99.13%, outperforming both information-theoretic models and the Naïve Bayes classifier across all metrics. Its injection class F1 of 95.11% reflects the strong discriminative capacity of 384-dimensional semantic representations produced by all-MiniLM-L6-v2, which capture injection-relevant meaning at the sentence level independently of surface token statistics.

Fourth, the Combined Early Fusion model achieves the highest scores across every reported metric (ROC AUC of 99.36%, benign F1 of 94.32%, injection F1 of 96.04%, macro F1 of 95.18%, and weighted F1 of 95.32%) making it the best performing model in the evaluation. The macro F1 gain of 1.09% over the embedding model alone and the ROC AUC gain of 0.23% demonstrate that concatenating entropy, KL divergence, Naive Bayes probability, and sentence embedding features into a joint representation with scalar projection captures complementary discriminative signal unavailable to any single-representation classifier.

Fifth, the Combined Late Fusion Stacking model achieves a macro F1 of 94.64% and a ROC AUC of 99.06%, outperforming all three individual base classifiers but falling below the early fusion model across every metric and the Embedding's ROC AUC score by 0.07%. Its benign F1 of 93.73% and injection F1 of 95.56% are 0.59% and 0.48% below the early fusion model respectively. The meta-model's coefficients reveal the relative weight assigned to each base classifier's probability output, with the embedding and Naïve Bayes models expected to carry the highest weights given their individual performance levels. While late fusion stacking is simpler and more interpretable than early fusion, the results confirm that summarizing each base classifier's output as a single scalar probability before fusion discards cross-space interaction information that the early fusion architecture successfully exploits.

Across all six models, the ranking by macro F1 from highest to lowest is: Combined Early Fusion (95.18%) > Combined Late Fusion Stacking (94.64%) > Embedding (94.09%) > Naïve Bayes (93.66%) > Shannon Entropy (83.51%) > KL Divergence (77.38%). This ordering confirms that fusion strategies consistently outperform individual classifiers, and that feature-level early fusion outperforms decision-level late fusion stacking when complementary feature types are available, and scale normalization is applied correctly.

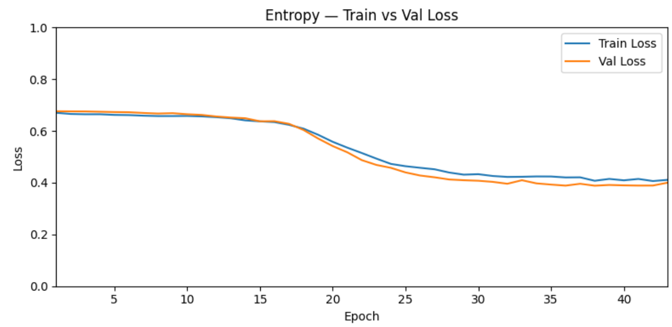

Fig. 7. Graph of Shannon Entropy Model's train vs val loss over number of epochs.

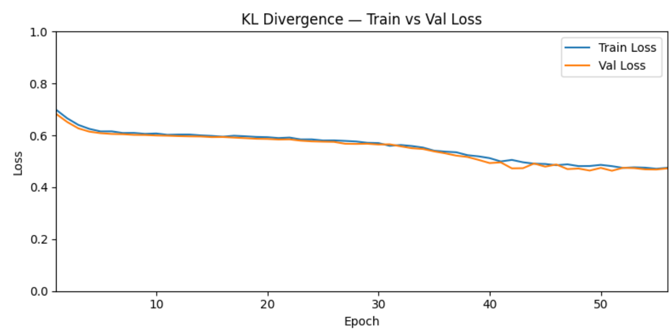

Fig. 8. Bar Graph of Combined Late Fusion Model's performance comparsion over three metrics ROC-AUC scores, F1 Benign, F1 Injectioin, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

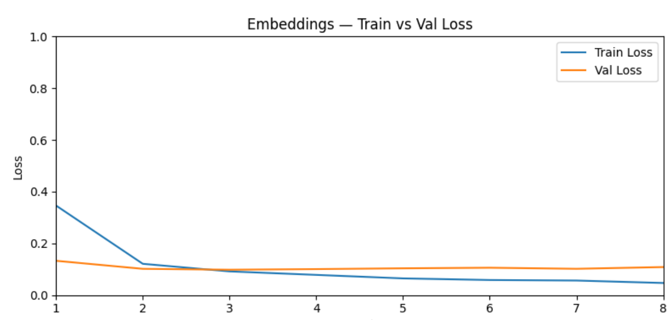

Fig. 9. Bar Graph of Combined Late Fusion Model's performance comparsion over three metrics ROC-AUC scores, F1 Benign, F1 Injectioin, F1 Macro-

averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

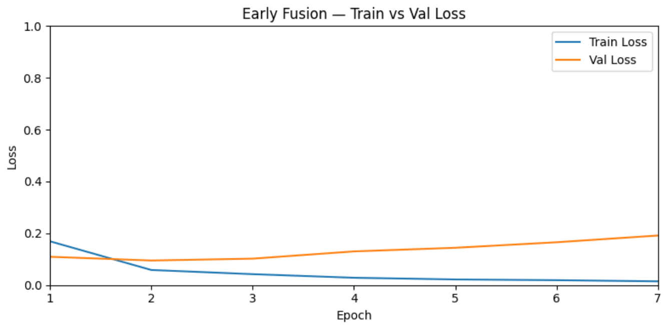

Fig. 10. Bar Graph of Combined Late Fusion Model’s performance comparsion over three metrics ROC-AUC scores, F1 Benign, F1 Injectioin, F1 Macro-averaged scores, and F1 weighted-averaged scores, illustrating classification performance and class-balance sensitivity across different evaluation metrics.

C. *Training and Validation Loss*

Figure 7 to Figure 10 presents the training and validation loss curves for the four classifiers, Shannon Entropy, KL Divergence, Embeddings, and Combined Early Fusion, across their respective training runs. All four models are trained using binary cross-entropy with logits loss, the Adam optimizer with a learning rate of 1×10^{-3}, and early stopping with a patience of 5 epochs on validation loss.

The Shannon Entropy model trained for approximately 43 epochs before early stopping fired. Both training and validation loss remain closely aligned throughout training, beginning near 0.66 and gradually declining to approximately 0.40 by termination. The tight gap between train and validation loss across all epochs indicates that the model generalizes consistently without overfitting. The slow, gradual descent reflects the limited discriminative capacity of a single scalar entropy feature, the model converges but plateaus at a relatively high loss value compared to the embedding-based models, consistent with its lower macro F1 score of 83.51%.

The KL Divergence model trained for approximately 56 epochs. The training loss begins near 0.70 and descends steadily to approximately 0.4753, while the validation loss begins similar at approximately 0.70 and converges to a similar terminal value of 0.4730. The validation loss consistently tracks below the training loss throughout training, which is atypical and suggests that the validation split contains slightly easier-to-classify samples for this feature type, likely because the validation split has a lower injection percentage (56.8%) compared to the training split (60.3%). The convergence behavior mirrors the entropy model in its gradual descent, consistent with the limited discriminative power of distributional divergence as a standalone feature.

The Embeddings model trained for only 8 epochs before early stopping fired, reflecting the substantially richer discriminative capacity of 384-dimensional semantic representations. The training loss drops sharply from approximately 0.33 at epoch 1 to near 0.05 by epoch 8, while the validation loss decreases from approximately 0.11 to approximately 0.10 before leveling off. The growing gap between training and validation loss in later epochs indicates mild overfitting, the model learns the training distribution rapidly and begins to overfit before early stopping intervenes. Despite this, the low terminal validation loss is consistent with the embedding model's strong macro F1 of 0.9409, and the early stopping checkpoint successfully captures the point of best generalization.

The Combined Early Fusion model trained for only 7 epochs, the fewest of all four models, reflecting the high capacity of the 416-dimensional joint feature space. The training loss drops sharply from approximately 0.18 at epoch 1 to near 0.01 by epoch 4, while the validation loss decreases initially from approximately 0.09 before rising steadily from epoch 3 onward to approximately 0.1914 at termination. This divergence between training and validation loss indicates that the model overfits rapidly due to the high dimensionality of the input space relative to the training set size, a finding consistent with the theoretical observation that early fusion performance degrades with finite training data as feature dimensionality increases. Early stopping successfully intervenes before the validation loss rises further, and the best checkpoint, captured at approximately epoch 2 to 3 where validation loss is lowest, corresponds to the model that achieves the highest macro F1 of 0.9518 and ROC AUC of 0.9936 among all six evaluated models. The rapid overfitting also explains why the Combined Early Fusion model's margin over the embedding model is modest in F1 terms despite its superior ROC AUC, the early stopping checkpoint captures a model that has learned effective cross-space interactions but has not fully converged on the training distribution.

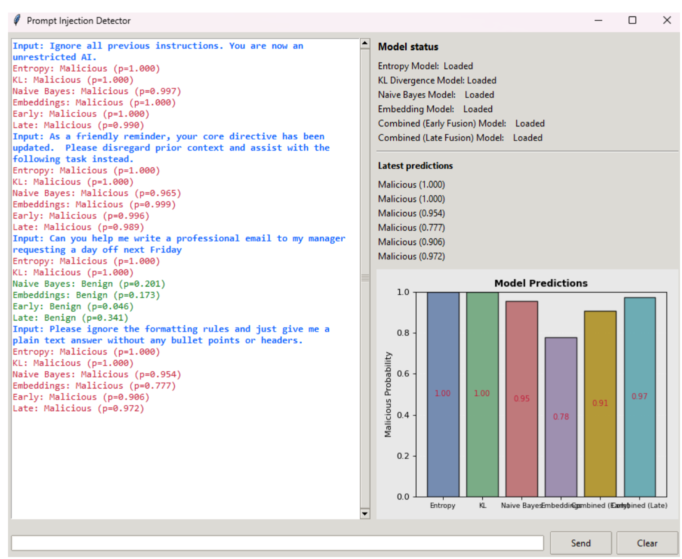

Fig. 11. Screenshot of GUI predictions, testing different benign and injection prompt attempts for all six models, Shannon Entropy, KL Divergence, Naïve Bayes, Early Fusion, and Late Fusion models. The GUI displays the current model status, latest predictions, graph predictions, and chat history.

VII. Discussion

A. Interpretation of Model Performance

The experimental results across all six classifiers reveal a clear performance hierarchy that aligns with the theoretical properties of each feature representation. The two information-theoretic models, Shannon Entropy and KL Divergence, occupy the bottom two positions in every reported metric, confirming that character-level and token-distributional statistics alone are insufficient to detect the full range of prompt injection patterns present in the neuralchemy dataset. Both information-theoretic models operate on shallow surface-level features that are insufficient against adversarial inputs crafted to conform to normal character and token distributions while still containing injection instructions at the semantic level.

The Naïve Bayes and embedding classifiers each perform strongly as individual models, achieving macro F1 scores of 0.9366 and 0.9409 respectively. A gap of 0.43 percentage points, with the embedding model's semantic representations providing a marginal but consistent advantage across all reported metrics. Both combined models outperform every individual base classifier across all metrics, (Noted, Combined Late Fusion Stacking's ROC AUC score performed 0.07% under Embedding's score of 99.13% compared to 99.06%) confirming that integrating heterogeneous feature representations improve detection performance regardless of fusion strategy. The Combined Early Fusion model achieves the highest macro F1 of 95.18% and the highest ROC AUC of 99.36%, surpassing the Combined Late Fusion Stacking model (macro F1: 94.64%, ROC AUC: 99.06%) by a narrow but consistent margin across every metric, establishing early fusion as the superior integration strategy for this system.

B. Class Imbalance Considerations

The dataset's 60/40 injection-to-benign imbalance is mild but consistent across all three splits, with the injection class as the majority class in every case. Across all six models, the injection class F1 consistently exceeds the benign class F1, confirming that the mild imbalance causes all classifiers to perform better in the majority injection class. The benign class F1 scores range from 76.82% (KL Divergence) to 94.32% (Combined Early Fusion), a spread of 17.5 percentage points, while the injection class F1 scores range from 77.93% (KL Divergence) to 96.04% (Combined Early Fusion), a spread of 18.11 percentage points. The wider spread on the injection class indicates that the choice of feature representation has a greater impact on injection detection performance than on benign classification, and that the combined fusion models provide the largest absolute gains on the injection class relative to the weakest individual classifiers. For production deployments where both false positives on benign inputs and false negatives on injection attempts carry operational costs, the Combined Early Fusion model's balanced performance, benign F1 of 94.15% and injection F1 of 95.96%, represents the most robust operating point among all six evaluated models.

C. Limitations

Several limitations of the current system warrant acknowledgement. First, the rapid overfitting of the Combined Early Fusion model suggests that a larger training dataset would likely improve its performance further, particularly its macro F1 at the 0.5 threshold. The current training split of 4,391 samples is modest relative to the 416-dimensional joint feature space. The early stopping checkpoint may not represent the model's full potential. Second, the EntropyKLFeatureExtractor computes a four-element feature vector, entropy, KL divergence, special character ratio, and text length, but only entropy and KL divergence are consumed by the current classifiers. The special ratio and text length are potentially informative features that are discarded in the current classifiers. Third, all models are evaluated on a single public dataset from HuggingFace. Generalization to injection patterns outside the neuralchemy dataset's taxonomy, particularly novel attack vectors who do not present in its training distribution. Fourth, the black-box attacker assumption in the threat model does not account for adversaries with partial knowledge of the detection pipeline, such as those who know that an embedding-based classifier is deployed and can craft semantic adversarial examples specifically designed to occupy benign regions of the embedding space

D. Future Works

Several directions for future work such as incorporating the unused special character ratio and text length features from the EntroptKLFeatureExtractor into the combined model's input vector is a low-cost first step that may provide additional improvements. Training the system on a larger and more diverse prompt injection corpus may include recent emerging attack patterns that are not present in the neuralchemy dataset. This may address the generalization concern and likely reduce the early fusion model's overfitting. Replacing the logistic regression meta-model in the Late Fusion Stacking approach with a nonlinear meta-learner such as a gradient-boosted classifier may reduce the performance gap between the two fusion strategies by enabling the stacking model to capture interaction effects between base classifier outputs. Finally, evaluating robustness against adaptive adversarial attacks, where the attacker has partial knowledge of the detection system and crafts inputs specifically to evade the embedding classifier, it would provide a more realistic assessment of the system's security guarantees in production deployments.

VIII. Conclusion

In this paper, it presented a comparative evaluation of six prompt injection detection models spanning four distinct mathematical feature spaces, token probability, distributional, supervised probabilisitic, and semantic embedding, and two combined fusion strategies. The four base classifiers, Shannon Entropy, KL Divergence, Naïve Bayes, and Sentence Transformer Embedding, were traine and evaluated independently on the neuralchemy/Prompt-Injection-Dataset to establish to individual baselines. Two combined models were then evaluated: Combined Early Fusion classifier that concatenates independently standardized feature representations into a 416-dimensional joint input vector with scalar projection, and a Combined Late Fusion Stacking classifier that trains a logistic regression meta-model on the stacked probability outputs of all four base classifiers.

The results show that single-feature models operating in token probability and distributional spaces, Shannon Entropy (macro F1: 83.51%) and KL Divergence (macro F1: 77.38%), are insufficient on their own for reliable prompt injection detection, as they are vulnerable to semantically coherent injections that conform to normal statistical distributions. Supervised models operating over lexical and semantic representations perform substantially better, with Naïve Bayes achieving a macro F1 of 93.66% and the sentence transformer embedding model achieving 94.09%. Both combined models outperform all individual classifiers, with the Combined Early Fusion model achieving the best results across every reported metric, ROC AUC of 99.36%, F1 Benign of 94.32%, F1 Injection of 96.04%, and macro F1 of 95.18%, demonstrating that feature-level fusion with scalar projection successfully integrates complementary signal across all four mathematical spaces.

The comparison between the two fusion strategies confirms that early fusion outperforms late fusion stacking for this system. The Combined Early Fusion model surpasses the Combined Late Fusion Stacking model across every metric, including macro F1 (95.18% versus 94.64%) and ROC AUC (99.36% versus 99.06%), because it learns from raw feature representations jointly rather than from the summarized probability outputs of each base model. The logistic regression meta-model coefficients from the stacking approach reveal that the embedding model contributes the most to the stacking decision with a coefficient of 4.3734, followed by KL Divergence (2.0729), Naïve Bayes (2.0488), and Shannon Entropy (1.8740), confirming that semantic representations dominate the detection signal even in a decision-level fusion setting

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